Topics Covered:

- Fping
- nmap

1. Find the network Configuration

After connecting to the connecting to the lab, check the network configuration of the TAP interface. Then use this information to configure your scans.

```
tap0      Link encap:Ethernet  HWaddr d6:b4:d8:c8:fe:d4        inet
     addr:10.142.111.240        Bcast:10.142.111.255
Mask:255.255.255.0           inet6 addr:

fe80::d4b4:d8ff:fec8:fed4/64 Scope:Link

UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1

          RX  packets:21025  errors:0  dropped:57  overruns:0  frame:0

TX    packets:49948    errors:0    dropped:0    overruns:0    carrier:0

collisions:0 txqueuelen:100

          RX bytes:8167465 (7.7 MiB)  TX bytes:3566191 (3.4 MiB)
```

2. Perform a Ping Scan with fping

Run a ping scan on the entire network with *fping*.

```
# fping  -a -g 10.142.111.0/24 2> /dev/null

10.142.111.1

10.142.111.6

10.142.111.48 10.142.111.96

10.142.111.99

10.142.111.100

10.142.111.240
```

Fping reports 6 hosts and our attacker machine.

3. Run a ping scan with NMAP

Running a ping scan with nmap reports 7 hosts. There is probably a host that does not respond to ICMP echo requests, but that has a service listening on the network.

```
Starting Nmap 6.47 ( http://nmap.org ) at 2015-02-23 18:51 CET Nmap scan report for 10.142.111.1

Host is up (0.18s latency).

MAC Address: 00:50:56:B1:E5:72 (VMware) Nmap scan report for 10.142.111.6

Host is up (0.19s latency).

MAC Address: 00:50:56:B1:02:7E (VMware) Nmap scan report for 10.142.111.48  Host is up

(0.20s latency).

MAC Address: 00:50:56:B1:16:C4 (VMware) Nmap scan report for 10.142.111.96  Host is up

(0.19s latency).

MAC Address: 00:50:56:B1:02:7E (VMware) Nmap scan report for 10.142.111.99  Host is up

(0.19s latency).

MAC Address: 00:50:56:B1:C1:0C (VMware) Nmap scan report for 10.142.111.100

Host is up (0.19s latency).

MAC Address: 00:50:56:B1:02:7E (VMware) Nmap scan report for 10.142.111.213  Host is up

(0.21s latency).

MAC Address: 00:50:56:B1:02:7E (VMware) Nmap scan report for 10.142.111.240  Host is up.

Nmap done: 256 IP addresses (8 hosts up) scanned in 5.01 seconds
```

4. Run A SYN Scan

This time run *nmap* only on the alive hosts.

```
# nmap -sS 10.142.111.1,6,48,96,99,100,213

Starting Nmap 6.47 ( http://nmap.org ) at 2015-02-23 18:51 CET Nmap scan report for 10.142.111.1
Host is up (0.18s latency).
Not shown: 997 filtered ports
PORT   STATE SERVICE
22/tcp open  ssh
53/tcp open  domain
80/tcp open  http
MAC Address: 00:50:56:B1:E5:72 (VMware)

Nmap scan report for 10.142.111.6  Host
is up (0.18s latency).
Not shown: 999 closed ports
PORT   STATE SERVICE
22/tcp open  ssh
MAC Address: 00:50:56:B1:02:7E (VMware)

Nmap scan report for 10.142.111.48  Host
is up (0.18s latency).
Not shown: 996 closed ports
```

```
PORT     STATE SERVICE
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
3389/tcp open  ms-wbt-server
MAC Address: 00:50:56:B1:16:C4 (VMware)

Nmap scan report for 10.142.111.96  Host
is up (0.19s latency).
Not shown: 999 closed ports
PORT   STATE SERVICE
80/tcp open  http
MAC Address: 00:50:56:B1:02:7E (VMware)

Nmap scan report for 10.142.111.99
```

```
80/tcp open  http
MAC Address: 00:50:56:B1:C1:0C (VMware)

Nmap scan report for 10.142.111.100  Host
is up (0.18s latency).
All 1000 scanned ports on 10.142.111.100 are closed
MAC Address: 00:50:56:B1:02:7E (VMware)

Nmap scan report for 10.142.111.213  Host
is up (0.18s latency).
Not shown: 999 closed ports
PORT   STATE SERVICE
81/tcp open  hosts2-ns MAC Address: 00:50:56:B1:02:7E (VMware)

Nmap done: 7 IP addresses (7 hosts up) scanned in 148.85 seconds
```

10.142.111.100 is probably a client as it does not listen on the network for connections.

5. Version Detection Scan

Run the version detection scan and spot services running on non-conventional default ports.

```
# nmap -sV 10.142.111.1,6,48,96,99,100,213

Starting Nmap 6.47 ( http://nmap.org ) at 2015-02-23 18:56 CET Nmap scan report for
10.142.111.1
Host is up (0.18s latency).
Not shown: 997 filtered ports
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 5.4p1 (FreeBSD 20100308; protocol 2.0)
53/tcp open  domain  dnsmasq 2.55
80/tcp open  http    lighttpd 1.4.29
MAC Address: 00:50:56:B1:E5:72 (VMware)
Service Info: OS: FreeBSD; CPE: cpe:/o:freebsd:freebsd

Nmap scan report for 10.142.111.6  Host
is up (0.18s latency).
Not shown: 999 closed ports
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 6.0p1 Debian 4+deb7u2 (protocol 2.0)
MAC Address: 00:50:56:B1:02:7E (VMware)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
Nmap scan report for 10.142.111.48  Host
is up (0.17s latency).
Not shown: 996 closed ports
```

```
Host is up (0.17s latency).
Not shown: 997 filtered ports
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 5.4p1 (FreeBSD 20100308; protocol 2.0)
53/tcp open  domain  dnsmasq 2.55
80/tcp open  http    lighttpd 1.4.29
MAC Address: 00:50:56:B1:C1:0C (VMware)
Service Info: OS: FreeBSD; CPE: cpe:/o:freebsd:freebsd
Nmap scan report for 10.142.111.100  Host
is up (0.17s latency).
All 1000 scanned ports on 10.142.111.100 are closed
MAC Address: 00:50:56:B1:02:7E (VMware)

Nmap scan report for 10.142.111.213  Host
is up (0.18s latency).
Not shown: 999 closed ports
PORT   STATE SERVICE VERSION 81/tcp open  http    Apache httpd 2.2.22 ((Debian))
MAC Address: 00:50:56:B1:02:7E (VMware)

Service detection performed. Please report any incorrect results at http://nmap.org/submit/ .
Nmap done: 7 IP addresses (7 hosts up) scanned in 181.57 seconds
```

10.142.111.213 runs Apache web server on a not standard port. Please note that this is the host which does not reply to ping echo requests.

6. OS Fingerprinting

Fingerprint the operating systems running on the hosts with the -O *nmap* option.

```
# nmap -O 10.142.111.1,6,48,96,99,100,213

Starting Nmap 6.47 ( http://nmap.org ) at 2015-02-23 19:27 CET Nmap scan report for 10.142.111.1

Host is up (0.19s latency).

Not shown: 997 filtered ports

PORT   STATE SERVICE

22/tcp open  ssh

53/tcp open  domain

80/tcp open  http

MAC Address: 00:50:56:B1:E5:72 (VMware)

Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port

Device type: general purpose|specialized|media device|broadband router

Running (JUST GUESSING): OpenBSD 4.X|3.X|5.X (92%), FreeBSD

Scientific Atlanta embedded (85%)

OS CPE: cpe:/o:openbsd:openbsd:4.3 cpe:/o:freebsd:freebsd:7.0

cpe:/o:freebsd:freebsd:9 cpe:/o:openbsd:openbsd:3 cpe:/o:openbsd:openbsd:4 cpe:/o:apple:iphone_os:5.2.1

cpe:/h:scientificatlanta:webstar_dpc2100

Aggressive OS guesses: OpenBSD 4.3 (92%), FreeBSD 7.0-RELEASE

(87%), FreeBSD 9.1-PRERELEASE (86%), Comau C4G robot control unit

(86%), OpenBSD 3.8 - 4.7 (85%), OpenBSD 4.1 (85%), OpenBSD 4.9 - 5.1 (85%)

No exact OS matches for host (test conditions non-ideal).

Network Distance: 1 hop
```

Exercises

1. Run a ping scan with fping
2. Run a ping scan with nmap, do you find any differences? Can you tell why?
3. Perform a SYN scan against the targets. Identify clients and servers.
4. Identify the version of every daemon listening on the network
5. Identify, if it is possible, the operating system running on each host.